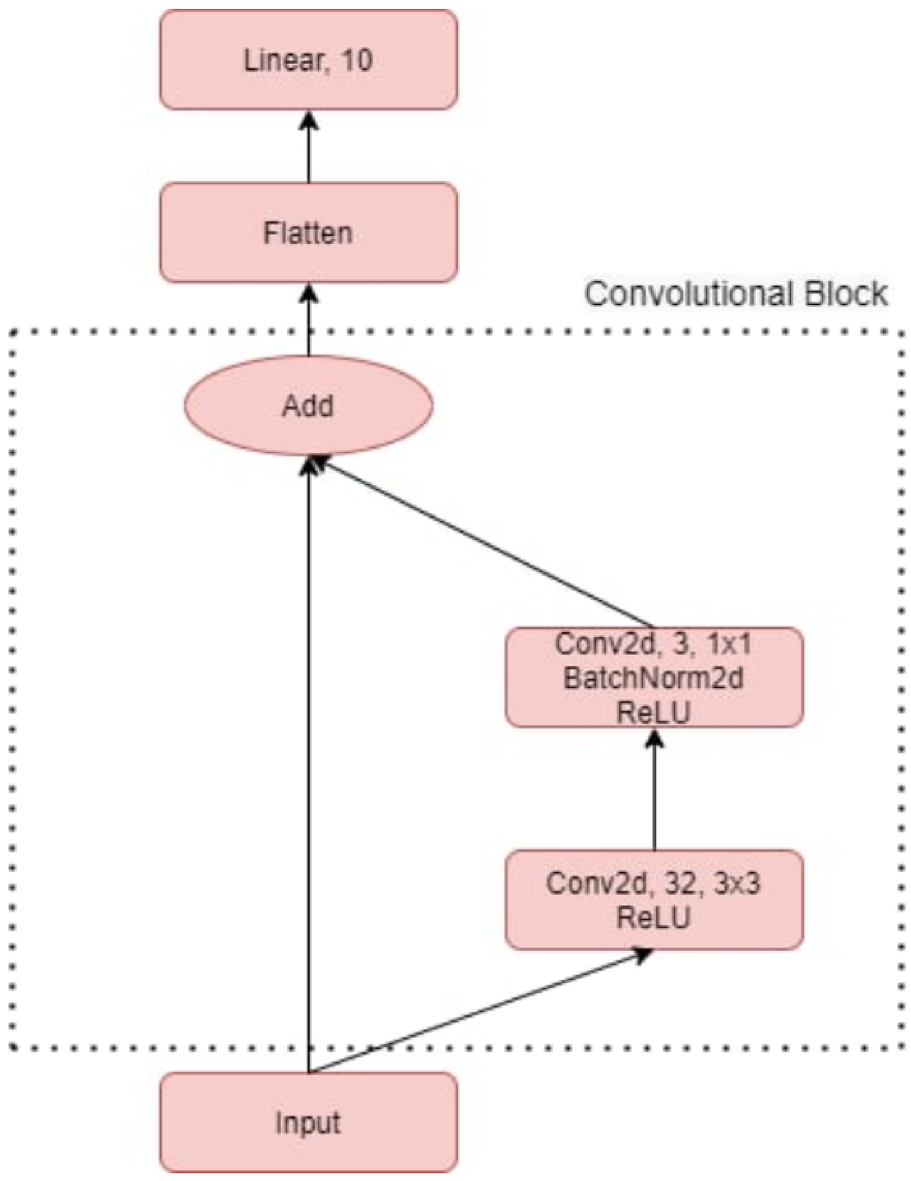
Laboratory 9 – Exercises

1. Define an input tensor of shape (1,1,7,7) with random values. The two-dimensional convolutional layer should have as arguments: $c_in = 1$, $c_out = 1$, $padding = (2, 1)$ and $stride = (3, 3)$. Set the *kernel_size* argument such that the output shape of the convolutional operation to be (1,1,3,2).

2. Classify the SVHN dataset (32×32 images, 10 classes, 73257 training images and 26032 testing images) using ResNet architecture by applying only two modules of residual blocks. Train the model for 10 epochs with $lr = 0.9$ and $batch_size = 256$. Plot the training and validation accuracy.

3. Classify the SVHN dataset (32×32 images, 10 classes, 73257 training images and 26032 testing images) using the following convolutional neural network architecture. Divide the training dataset as follows: 30000 validation images and 43257 training images. Train your model for 5 epochs using a learning rate of 0.05 and a batch size of 256. Define a class for implementing the Convolutional Block. Conv2d, 32, 3×3 means 32 output channels and 3×3 kernel size. Use padding, if necessary.